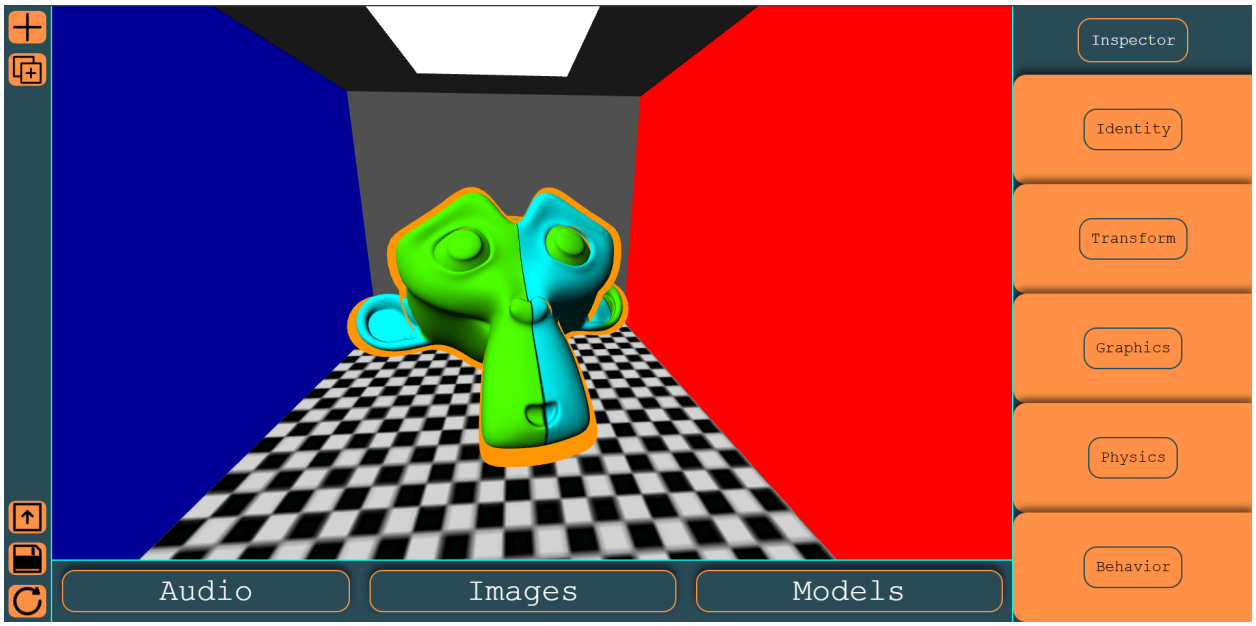
ARC-Engine User Guide

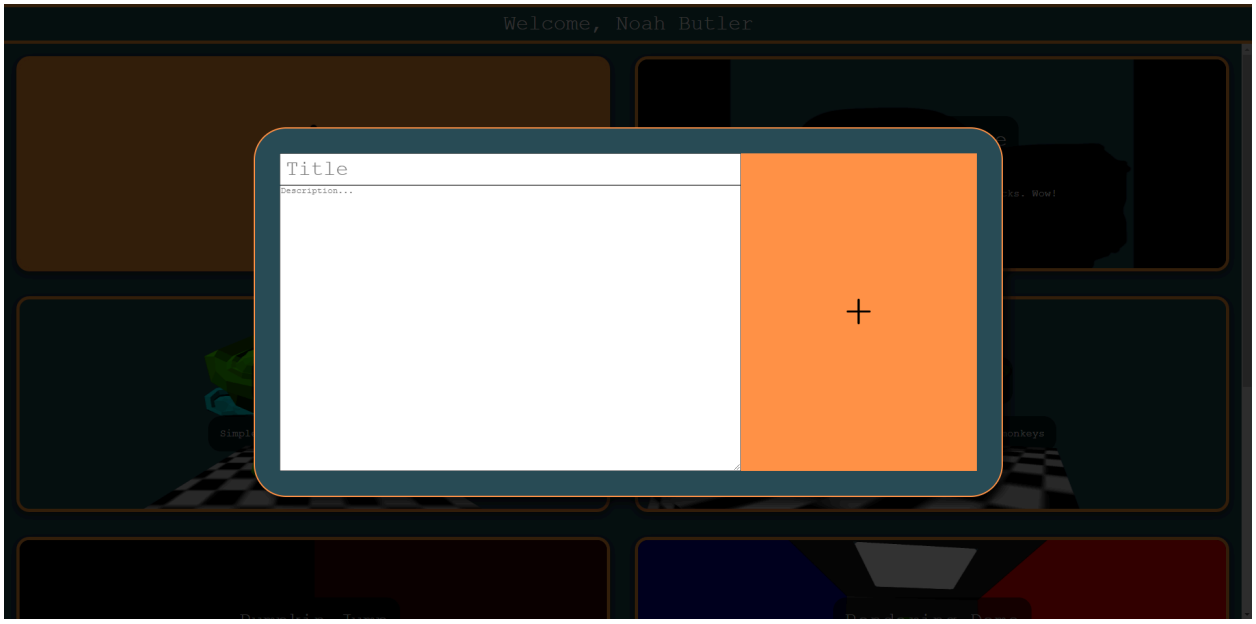
Noah Butler



Project Setup

- Download and install the most recent LTS version of Node.Js "https://nodejs.org/en"
- Open up a terminal window in the project's directory and run "npm i" to install the project's dependencies.
- Run "node server.js" in the same terminal window to run the server on your machine.
- Visit "http://localhost/" in your browser of choice to begin interacting with the project.

Site Navigation

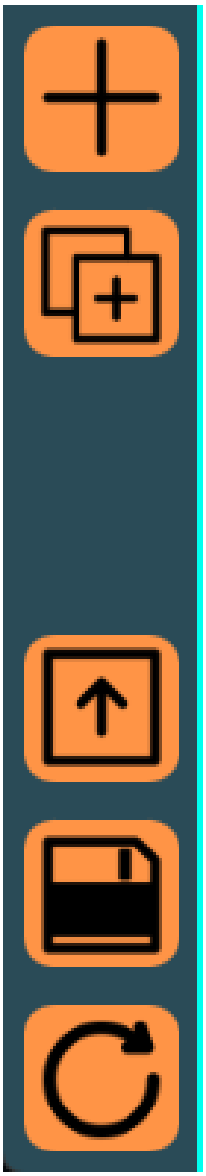


- On the home page, click “Login” to be brought to a Google OAuthorization page. Login with your Google account to be brought to your profile page.
- Click the orange tile with a “+” symbol on it to open up a menu where you can begin initializing a new game. Fill out the fields for project name and description. Once you are finished, click the “+” button on the right to create a new game and add it to your profile page as a tile.
- Click on the tile you just created (or select another project if you have any) to be brought to the game engine page where you can begin creating your new game.

At any time while on the engine page, press Ctrl+P to enter playmode and begin playing your game. Pressing Ctrl+P again will exit playmode and bring you back to the engine page view.

Engine Usage

- The toolbar on the left contains a series of buttons that allow you to manipulate the game's contents. Below is a list ordered by appearance (top to bottom) explaining each tool:



- The plus button allows you to create a new entity in the scene. Clicking the button will add the new entity to the scene list on the right and display it in front of the camera in the center viewport.

- The copy button will duplicate the entity you have currently selected in the scene list. Clicking the button will create a new copy of the selected entity with all of the same attributes and properties.

- The upload button will open up a file select menu where you can upload an asset to the engine from your computer. The allowed file types are .png, .jpg, .jpeg, .mp3, .wav, and .json (models must be uploaded in json format).

- The save button allows you to save the current state of your game, including all of the entities in the scene.

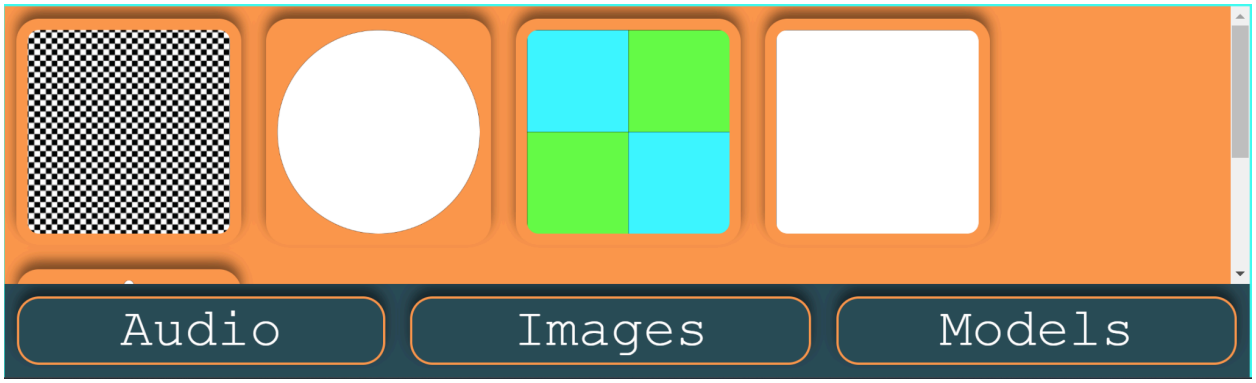
- The discard button at the very bottom allows you to discard all of the changes you have made since your last save, reverting you back to the last version.



- The viewport in the center displays a rendering of the scene, including the entities within it. After clicking the plus button in the toolbar, a cube will be added to the scene and appear in the viewport. To traverse the viewport and view the cube from multiple angles, the following controls are available:
    - Left mouse button: Press and drag your mouse to move around the viewport on a 2D axis (up, down, left, right).
    - Right mouse button: Press and drag your mouse to look around the viewport in 3D space.
    - Middle mouse button: Press and drag your mouse to pan around a point one meter in front of the camera.
    - Scroll wheel: Scroll up to move the camera forwards in the direction you are facing. Scroll down to move backwards instead.
    - W, A, S, D, Q, and E keys: For more precise movement while pressing the right mouse button to look around, hold down the W key to move forwards, the S key to move backwards, the A key to move leftwards, the D key to move rightwards, the Q key to move downwards, or the E key to move upwards.



- The inspector on the right displays a list of all the objects currently in the scene. After creating an object using the plus button in the toolbar, an entry will appear in this list. Clicking on one of the entries in the list will select the entity, add an orange outline to it in the viewport, and change the inspector to display a collection of orange tabs, each one revealing the attributes of the selected entity when clicked on. Clicking the “Inspector” button at the top will deselect the current entity and swap the inspector to display the scene list again.



- The taskbar at the bottom of the screen (below the viewport) displays buttons representing the different folders that contain all of the assets available to the project. Clicking on each button pops out a drawer containing the individual assets. While an entity is selected, opening the “Images” drawer and selecting an image asset will apply the image as a texture for the entity. Similarly, opening the “Models” drawer and selecting a model asset will swap the selected entity’s model to the one chosen.



- The terminal is a window accessible by clicking the “Behavior” tab that appears in the inspector when an entity is selected. Clicking this tab will open a terminal containing a text box containing sample code with a few empty functions. To the right of the text box is a list of attributes and methods accessible to you for reference while you are writing a program for the selected entity. After writing your own code in the text box, simply click the “Behavior” tab again to close the terminal. The sample methods behave as follows:
    - Start(): This method is called during the first frame that the game runs.
    - Update(): This method is called every frame while the game is running.
    - KeyPress(key): This method is called every time the player presses a key on their keyboard. The argument passed in for “key” is the lowercase representation of the keyboard character they pressed.
    - Collide(other): This method is called every time the entity containing the script collides with another entity. The argument passed in for “other” is the entity that it collided with.